

Project Name	RTDIP Timeseries Forecasting
Online team meeting	https://fau.zoom-x.de/j/65502405196?pwd=8H73lyixQfqKrnO8Eb47kawnuKCnGp.1
Production system (if any)	
Test system (if any)	
GitHub repository	https://github.com/amosproj/amos2025ws03-rtdip-timeseries-forecasting
GitHub feature board	https://github.com/orgs/amosproj/projects/91/
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/96
Team T-shirt (white)	https://www.shirtinator.de/s/QWcDXTKGS72ISnrRNZq1hg
Team T-shirt (black)	https://www.shirtinator.de/s/0yx3duRjSzW-5hP32KmZ1w
Additional materials	https://discord.gg/KJXGmjcs
Team mailing list	oss-amos-projX@lists.fau.de
Happines Index App	https://happy-amos.appspot.com/Courses
Planning Poker	https://planningpokeronline.com/DMhnF5cOAK9ffF5jsRf2/

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#	Meeting Day	Product Owner		Software Developer	Release Manager	Scrum Master	Comment
		Review	Planning				
1	2025-10-15	Patrick Meusling	Hafidz Arifin	Everyone else	N/A	Luca	
2	2025-10-22	Hafidz Arifin	Patrick Meusling	Everyone else	Christoph	Luca	
3	2025-10-29	Patrick Meusling	Hafidz Arifin	Everyone else	Mehdi Khabouze	Luca	
4	2025-11-05	Hafidz Arifin	Patrick Meusling	Everyone else	Rene Jokiel	Luca	
5	2025-11-12	Patrick Meusling	Hafidz Arifin	Everyone else	Abdul Haseeb	Luca	
6	2025-11-19	Hafidz Arifin	Patrick Meusling	Everyone else	Hannes Pohnke	Luca	
7	2025-11-26	Patrick Meusling	Hafidz Arifin	Everyone else	Simon Selbig	Luca	Mid-term due
8	2025-12-03	Hafidz Arifin	Patrick Meusling	Everyone else	Rene Jokiel	Luca	
9	2025-12-10	Patrick Meusling	Hafidz Arifin	Everyone else	Mehdi Khabouze	Luca	
10	2025-12-17	Hafidz Arifin	Patrick Meusling	Everyone else	Christoph Huy	Luca	
11	2026-01-07	Patrick Meusling	Hafidz Arifin	Everyone else	Hannes Pohnke	Luca	
12	2026-01-14	Hafidz Arifin	Patrick Meusling	Everyone else	Simon Selbig	Luca	
13	2026-01-21	Patrick Meusling	Hafidz Arifin	Everyone else	Mehdi Khabouze	Luca	
14	2026-01-28	Hafidz Arifin	Patrick Meusling	Everyone else	Hannes Pohnke	Luca	
15	2026-02-04	Patrick Meusling	Hafidz Arifin	Everyone else	Christoph Huy	Luca	Demo day!
16	2026-02-11						Retrospective
Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

Goals	Aquire new skills
	Produce a functioning and valuable product (a 1.0)
Meeting norms	We respect the opinions of others and we show up on time
	Summary of Partner Meeting required
Working norms	Clean, testable code with clear commit messages, code comments
	We respect other people's work
Coordination norms	Seperate branches for issues, second person review
	We balance workload among the team
Communication norms	Daily check the corresponding channels (discord)
	We communicate constructively, make sure to communicate possible problems (absence, etc.)
Consideration norms	We discuss issues openly
	We vote in case we can't reach a consensus
Cont. improvement norms	We encourage critique and improvement efforts
	Encourage reaching out for help (use the strength/skills of others)
Rewards	We treat ourselves to a sweet of choice for good work
Sanctions	Push Ups (amount decided on the case)
Signatures	
Scrum Master	Luca Böhm
Product owner	Patrick Meusling
Product owner	Hafidz Arifin
Software developer	Abdul Haseeb
Software developer	Rene Jokiel
Software developer	Simon Selbig
Software developer	Hannes Pohnke
Software developer	Christoph Huy
Software developer	Mehdi Khabouze

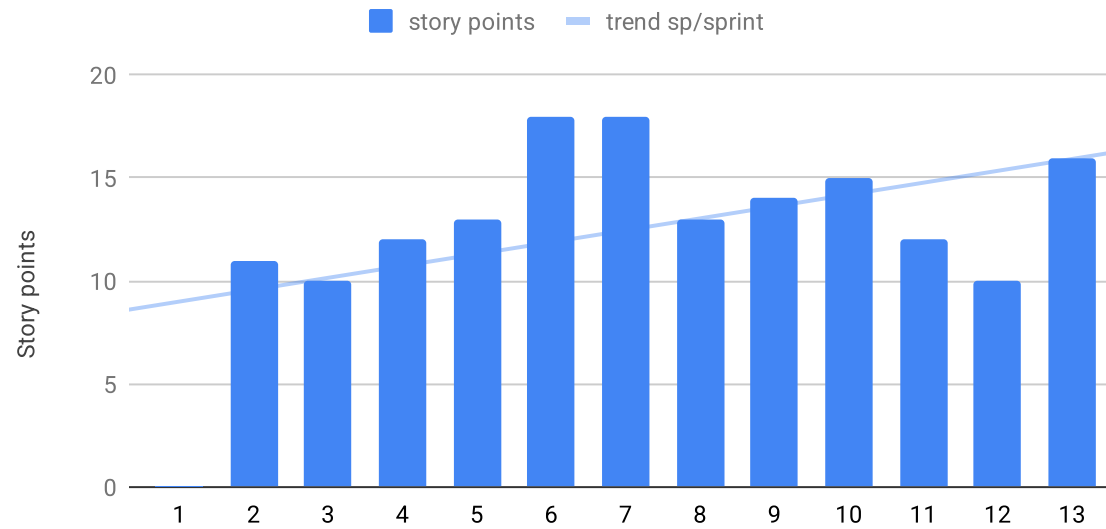
Product Vision	Project Mission
To empower organizations with open-source, scalable, and transparent forecasting capabilities that enable data-driven decision-making across industries. By extending RTDIP with advanced forecasting and anomaly detection components as well as sector specific datasets, we envision a future where time series insights such as trends, seasonality, and predictive patterns are seamlessly integrated into real-time data pipelines. This will allow businesses to understand historical data, anticipate future behavior, optimize operations, and improve efficiency using accessible, community-driven, and production-ready forecasting tools.	To design, develop, and contribute open-source forecasting components for the RTDIP platform that enable trend analysis, anomaly detection, and predictive modeling on time series data. Our mission within this project is to research and implement forecasting techniques using Python and Apache Spark, validate and enrich them with real-world datasets, and ensure they meet RTDIP's modular, tested, and well-documented standards. By doing so, we will enhance RTDIP's functionality and provide the open-source community and industry users such as Shell with reliable, scalable forecasting capabilities that integrate seamlessly into existing data pipelines.

Term	Definition
RTDIP (Real-Time Data Ingestion Platform)	An open-source project under the Linux Foundation designed to simplify the ingestion, transformation, and storage of data from various sources using scalable cloud technologies.
Forecasting	The process of predicting future values or trends based on historical time series data.
Time Series Data	Data collected over time at regular intervals, often used to analyze patterns such as trends and seasonality.
Apache Spark	An open-source distributed computing system that provides fast data processing for large-scale datasets.
PySpark	The Python API for Apache Spark, allowing developers to write Spark applications using Python.
Delta Lake	A storage layer that brings reliability, consistency, and performance to data lakes by supporting ACID transactions and schema enforcement.
Anomaly Detection	The identification of unusual patterns or outliers in data that do not conform to expected behavior.
Predictive Modeling	A statistical or machine learning approach that uses historical data to predict future outcomes.
Time Series Decomposition	The process of breaking down a time series into its core components such as trend, seasonality, and residuals.
ETL (Extract, Transform, Load)	A data pipeline process that extracts data from sources, transforms it into a suitable format, and loads it into a storage system.
Open Source	Software that is freely available to use, modify, and distribute, typically developed collaboratively by a community.
Linux Foundation	A nonprofit organization that supports open-source software projects and fosters collaboration across industries.
EasyCLA (Easy Contributor License Agreement)	A system used by the Linux Foundation to manage contributor license agreements, ensuring legal compliance for open-source contributions.
Modular Architecture	A software design principle that divides a system into separate, interchangeable components that can be developed and maintained independently.
Unit Testing	The practice of testing individual units or components of software to ensure they work as intended.
Documentation	Written descriptions and guides that explain the design, usage, and functionality of a software system.
Data Pipeline	A set of processes that move, transform, and store data from one system to another in a structured and automated way.
Scalability	The ability of a system to handle increasing amounts of work or data by adding resources.
Contribution Guidelines	A set of rules and best practices that contributors must follow to ensure consistency and quality in open-source projects.

Sprint #	Sprint goal
1	None
2	None
3	None
4	Optional
5	Establish a working foundation for the forecasting pipeline
6	Integrate the pipeline into the existing software
7	Advance the pipeline
8	Add more Datasets to the pipeline
9	Testing and Analysing the new Datasets
10	Adding new and different models to the pipeline
11	Testing and Analysing the new Models
12	Finalize the Product
13	Finalize for Demo day
14	Merging the product with the partner, coordinating the merge (they said this might take a bit of time)
15	Backup Sprint

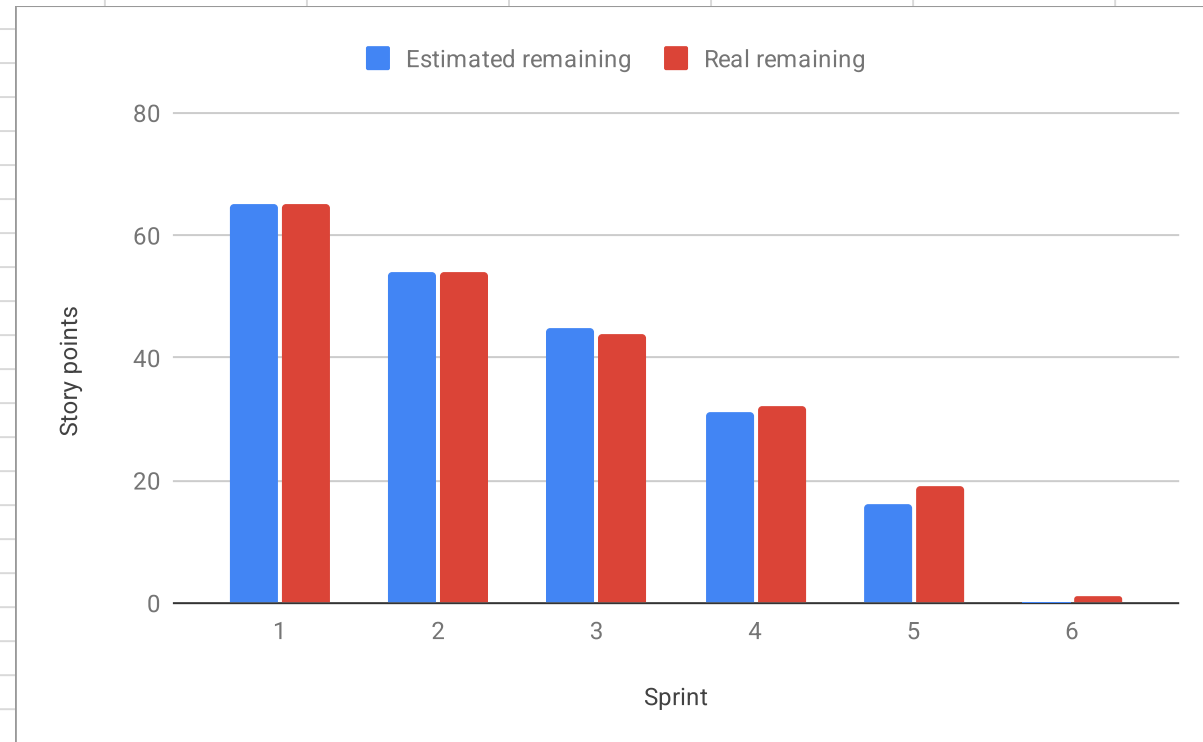
Sprint #	Story Points Realized
1	0
2	11
3	10
4	12
5	13
6	18
7	18
8	13
9	14
10	15
11	12
12	10
13	16
14	7
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB

Team 3 Velocity Chart



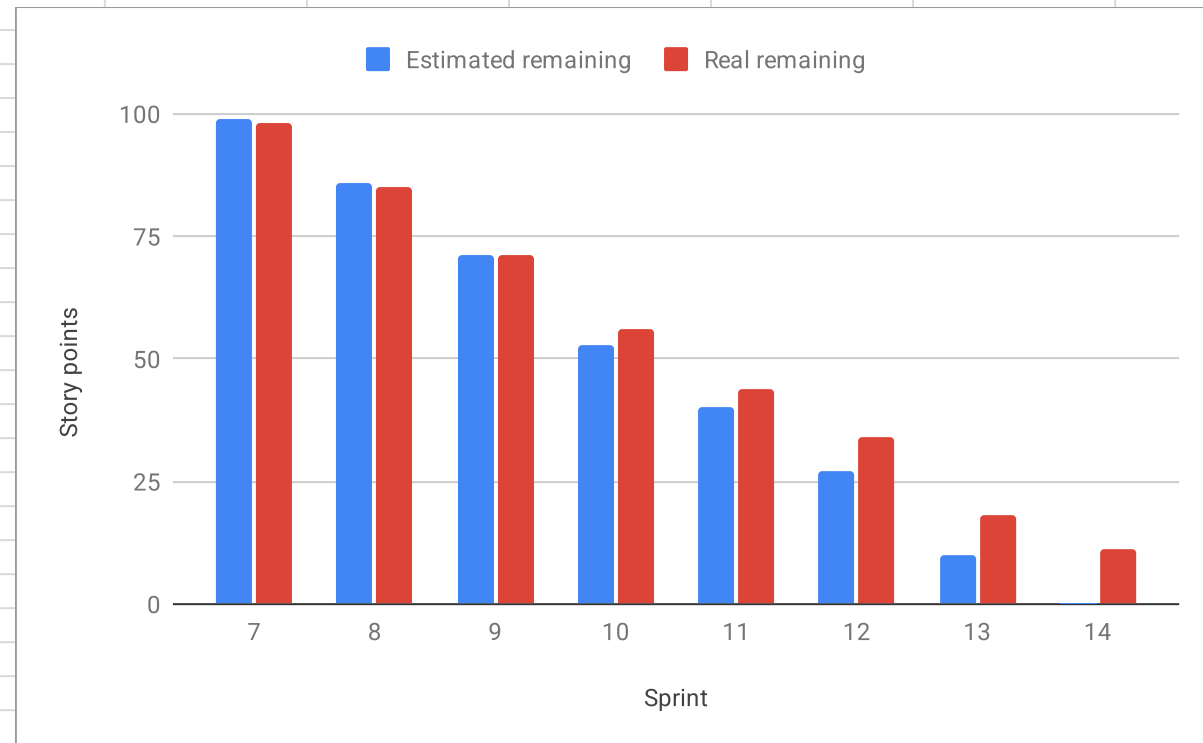
Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			65	65		
Sprints						
1	None		0	65	0	65
2	None		11	54	11	54
3	None		9	45	10	44
4	Optional		14	31	12	32
5	Establish a working foundation for the forecasting pipeline		15	16	13	19
6	Integrate the pipeline into the existing software		16	0	18	1
Features						
1	None					
2	None					
		Run the software on your own machine	1		1	
		Understanding the rtdip software	2		2	
		Software Architecture	2		2	
		Find open source time series datasets	2		2	
		Initialize software bill of materials	2		2	
		Research about time series forecasting fundamentals	2		2	
3	None					
		Research Possible Models Useful for Shell Dataset	2		2	
		Update our Forked Repository with Latest RTDIP Release	1		1	
		Prepare and Demo Project Build Process	1		2	
		Perform Data Exploration on Shell Dataset	3		3	
		Research about time series forecasting fundamentals	2		2	
4	Optional					
		Getting to Know HPC of AMOS	2		Dropped from Backlog	
		Perform Data Exploration on Opensource Dataset 1	2		3	
		Preprocess Shell Dataset	3		3	
		Implementing a basic Model pipeline	2		2	
		Research Possible Models Useful fo Opensource Dataset	2		2	
		CI/CD Pipeline fail bug	3		2	
5	Establish a working foundation for the forecasting pipeline					
		Choosing Models, Datasets, Error Measure Metrics	2		2	
		Create Build Process Video	2		2	
		Model Training and Evaluation with Shell Dataset	3		3	
		Visualization of Results	2		1	
		Preprocess Opensource Dataset 1	3		3	
		PO Homework	3		2	

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Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			116	116		
Sprints						
7	Advance the pipeline		17	99	18	98
8	Add more Datasets to the pipeline		13	86	13	85
9	Different Methods to Improve Forecast Accuracy		15	71	14	71
10	Customer Feedback and Adjustments		18	53	15	56
11	Improving Forecast Accuracy		13	40	12	44
12	Finish the Product		13	27	10	34
13	Demo Day Preparation		17	10	16	18
14	Finish the Project		10	0	7	11
Features						
7	Advance the pipeline					
		Collect and Document Customer Feedback on Project Progress	1		1	
		Adjust and Fix CI/CD Pipeline	3		3	
		Model Optimization and Improvement	3		3	
		Implementation of Time Series Decomposition	3		3	
		Implementation of Anomaly Detection	5		5	
		Model Training and Evaluation of Dataset SCADA	2		3	
8	Add more Datasets to the pipeline					
		Perform Data Exploration additional Opensource Dataset	3		3	
		Implement RTDIP Ingestion	2		2	
		Restructure and Refactor Wiki	2		2	
		Standardize Visualization	3		3	
		Model Optimization and Improvement for Timeseries Forecasting	3		3	
9	Different Methods to Improve Forecast Accuracy					
		Optimization and Improvement of Anomaly Detection Approach	3		3	
		Timeseries Forecasting Improvement with Other Models (Person 1)	3		3	
		Integrate Timeseries Forecasting Results Visualization into the SDK	3		3	
		Apply Timeseries Decomposition on shell Dataset	3		2	
		Timeseries Forecasting Improvement with Other Models (Person 2)	3		3	
10	Customer Feedback and Adjustments					
		Refactor Components for Spark-Compatible API	3		2	
		Document previous features in the software architecture	3		2	
		Adjusting previous features to base enviroment	2		1	
		Base Enviroment	8		8	
		Add Visualiualization Component for Anomaly Detection	2		2	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
11	Improving Forecast Accuracy					
		SCADA Decomposition	3		3	
		Refactor Wiki	2		2	
		Add Visualization Component for Timeseries Decomposition	2		2	
		Preprocessing of 3W_Dataset	3		3	
		Assesment of Robustness and long time predictions	3		2	
12	Finish the Product					
		Anomaly Detection and Evaluating 3W_Dataset	2		2	
		Training, Forecasting and Evaluating 3W_Dataset	3		3	
		Make all Datasets Running with All Models and Evaluate All	3		2	
		Merge Preparation	5		3	
13	Demo Day Preparation					
		Create demo day slide	1		2	
		Create demo video	3		3	
		Merge Project to Core Repository and Adjust based on Shell Feedback	5		5	
		Review Possible Fixes for Import Issue	5		3	
		Create Demo Day Presenation	3		3	
14	Finish the Project					
		Merge Project to Core Repository based on Shell Feedback	3		1	
		Project Report	1		1	
		Preparing and showcasing Implementation on Demo Day	3		3	
		Finalize user, (technical) design, and build/deploy documentation	3		2	
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				



#	Feature Definition of Done	Sprint Release Definition of Done	Project Release Definition of Done
	Acceptance criteria are met.		
	Work are pushed to the Github repository.		
	Create a branch for each backlog items (coding)		
	A pull request is created for each related branch.		
	The work products in the pull requests are reviewed.		
	The corresponding branches are merged and closed.		
	The bill of materials section of the planning documents is updated.		
	Work needs to be documented in the corresponding wiki section		
	For new features unit test have to be written.		
	Update our forked repository with the latest RTDIP release.		
	If the task involves coding, the implementation is integrated into the RTDIP framework and verified to function correctly within it.		
	All unit tests must pass successfully in the continuous integration (CI) pipeline.		
	All assignees must communicate and coordinate with each other to complete the task.		
		Release candidates with a working and meaningful update to the previous sprint is tagged.	
		Previously established features must continue to work.	
			The RTDIP forecasting components can be successfully built and deployed within the RTDIP environment.
			All automated tests and validation checks for the developed components pass without errors.
			The implemented forecasting features have been tested end-to-end and verified through a basic user workflow.
			Developer documentation is complete, clearly describing the architecture, setup, and contribution process for RTDIP.
			User documentation (usage examples, configuration steps, expected outputs) is finalized and up to date.
			The final release has been reviewed and approved by all project team members and the industry partner (Shell).

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Last Name	First Name	Value		#DIV/0!	#DIV/0!		
Böhm	Luca						
Meusling	Patrick						
Arifin	Hafidz						
Huy	Christoph						
Selbig	Simon			0	No size		
Haseeb	Abdul			1	Trivial size		
Pohnke	Hannes			2	Small size		
Jokiel	Rene			3	Medium size		
Khabouze	Mehdi			5	Large size		
				8	Very large size		
				13	Too large (size)		
How to play planning poker							
1. Everyone type their number into their value field, don't hit return yet							
2. Someone, perhaps a product owner, count down 3.. 2.. 1..							
3. Then, everyone hit return to submit their value							